

Real-Time Embedded Guidance and Reinforcement Learning for Autonomous Fixed-Wing UAV Interception of a Moving Target: A Hardware-in-the-Loop Validation Approach

Project description

The objective of this project is to develop and validate an autonomous interception system for a fixed-wing UAV using both classical guidance laws and Reinforcement Learning (RL). The interceptor must autonomously pursue and intercept a moving aerial target while respecting the kinematic constraints of fixed-wing aircraft. The developed algorithms will be deployed on embedded hardware and evaluated through a Hardware-in-the-Loop (HIL) simulation platform.

The project combines real-time embedded systems, autonomous decision-making, machine learning, and UAV guidance and control.

Objectives

The project aims to:

- Develop a kinematic model of a fixed-wing interceptor and a moving target.
- Implement conventional interception guidance laws.
- Develop a Reinforcement Learning-based guidance agent.
- Compare learning-based and model-based interception strategies.
- Deploy the guidance system on embedded hardware.
- Validate the complete system using HIL simulation.

Problem Statement

A fixed-wing UAV must autonomously intercept a moving aerial target. The UAV receives:

- Relative target position
- Relative target velocity
- Own aircraft state

and must generate guidance commands that minimize:

- Interception time
- Miss distance
- Energy consumption

while satisfying aircraft maneuverability constraints.

System Architecture

Layer 1: State Estimation

Provides:

- Relative position
- Relative velocity
- Line-of-sight angle
- Closing velocity

Three approaches can be used :

- GPS-based tracking → easiest for developing interception algorithms.
- Vision-based tracking → more realistic for non-cooperative targets.
- Sensor fusion (GPS + Vision + Lidar) → advanced research-level solution.

Layer 2: Guidance System

Two approaches will be implemented.

- Classical Guidance
 - Pure Pursuit
 - Lead Pursuit
 - Proportional Navigation (PN)
 - Augmented PN

These serve as performance baselines.

- Reinforcement Learning Guidance

The RL agent learns interception strategies through repeated interactions with the simulation environment. Possible algorithms:

- Deep Q Network (DQN)
- Proximal Policy Optimization (PPO)
- Soft Actor-Critic (SAC)

PPO is particularly attractive because it is stable and widely used for UAV control.

Layer 3: Flight Control

Low-level controllers:

- Roll control
- Pitch control
- Heading control
- Airspeed control

Implemented on:

- PX4
- STM32-based autopilot

Training Environment → Training occurs entirely in simulation.

Possible tools:

- PX4 SITL
- Gazebo
- AirSim

The Reinforcement Learning agent experiences thousands of interception scenarios.

Hardware-in-the-Loop Validation

After training:

1. The neural network is exported.
2. The policy runs on embedded hardware.
3. Real autopilot hardware is connected to the simulator.

HIL components:

- Pixhawk flight controller
- Raspberry Pi 5 or Jetson Orin Nano
- PX4 SITL
- Gazebo

Experimental Scenarios

- Scenario 1 : Straight-line target
- Scenario 2 : Fast moving target
- Scenario 3 : Target with heading changes
- Scenario 4 : Maneuvering target

Performance Evaluation

The RL approach will be compared with:

- Pure Pursuit
- Lead Pursuit
- Proportional Navigation
- Augmented PN

Metrics:

Metric	Description
Success Rate	Percentage of interceptions
Interception Time	Time to capture
Miss Distance	Final interception error
Energy Consumption	Estimated control effort
CPU Utilization	Embedded execution load
Inference Time	Neural network execution time
Memory Footprint	RAM usage

Expected Contributions

Scientific Contributions

- Reinforcement Learning formulation for fixed-wing UAV interception.
- Comparative analysis between RL and classical guidance laws.
- Study of generalization to unseen target trajectories.
- Real-time deployment of learned guidance policies.

Engineering Contributions

- Embedded interception software.
- Trained RL guidance agent.
- Hardware-in-the-Loop validation framework.
- Benchmark dataset of interception scenarios.

This topic naturally combines:

- Embedded Systems
- Real-Time Computing
- Reinforcement Learning
- Guidance and Control
- Fixed-Wing UAVs
- Hardware-in-the-Loop Simulation

REFERENCES

- ZARCHAN, Paul. Tactical and Strategic Missile Guidance. 7th ed. Reston, VA: American Institute of Aeronautics and Astronautics (AIAA), 2012.
- BAR-SHALOM, Yaakov, LI, Xiao-Rong and KIRUBARAJAN, Thiagalingam. Estimation with Applications to Tracking and Navigation. New York: John Wiley & Sons, 2001.
- ZHAO, S., SHAO, Z. and LIU, H. Evaluation of Proportional Navigation for Multirotor Pursuit. Drones, 2021, vol. 5, no. 1.
- VRBA, Matěj, et al. Planning for Autonomous Aerial Interception of UAVs. In: IEEE International Conference on Robotics and Automation (ICRA). Paris, France: IEEE, 2020.

- GIANNI, Nicolò, et al. Towards Safe Mid-Air Drone Interception. arXiv preprint arXiv:2405.13542, 2024.
- CHEN, Y., ZHANG, H. and LI, X. Multi-UAV Pursuit-Evasion with Online Planning by Deep Reinforcement Learning. arXiv preprint arXiv:2409.15866, 2024.
- LIU, Y., WANG, J. and ZHANG, X. A Dual Curriculum Learning Framework for Multi-UAV Pursuit-Evasion. arXiv preprint arXiv:2312.12255, 2023.
- YANG, K., LI, P. and ZHOU, T. Deep Reinforcement Learning-Based Intelligent Maneuvering Strategy for UAV Pursuit-Evasion. Drones, 2024, vol. 8, no. 7, article 309.
- PHAM, H., et al. Game of Drones: Multi-UAV Pursuit-Evasion Game with Online Motion Planning by Deep Reinforcement Learning. Aerospace, 2022, vol. 9, no. 2.
- BRISSET, Pascal, DOUARIN, Christophe, GORRA, François and DELAHAYE, Daniel. The Paparazzi Solution. In: MAV Conference. Braunschweig, Germany, 2006.
- DUTTA, R. A Hardware-in-the-Loop Simulator for UAV Autopilot Research. arXiv preprint arXiv:0804.3874, 2008.
- GANS, N., DIXON, W.E., LIND, R. and HENRY, D. Hardware-in-the-Loop Simulation for Vision-Based UAV Control. Journal of Intelligent and Robotic Systems, 2009, vol. 54, no. 1–3, pp. 255–273.
- KASNKOGLU, C. and TANRIKULU, O. Hardware-in-the-Loop Test Platform for Small Fixed-Wing UAVs. In: International Conference on Unmanned Aircraft Systems (ICUAS). Atlanta, USA: IEEE, 2013.